1. Use mathematical induction to prove:

$$\sum_{i=1}^{n} \frac{1}{i(i+1)} = \frac{n}{n+1}, \quad n \geq 1$$

① Base case, $n = 1$ (verify)

$\sum_{i=1}^{1} \frac{1}{i(i+1)} = \frac{1}{1+1}$ LHS: $\frac{1}{1(1+1)} = \frac{1}{1(2)} = \frac{1}{2}$ RHS: $\frac{1}{1+1} = \frac{1}{2}$ $\frac{1}{2} = \frac{1}{2}$

verified

② Assume that the following is true for $n = k$

$$\sum_{i=1}^{k} \frac{1}{i(i+1)} = \frac{k}{k+1}$$

③ Target for $n = k+1$

$$\sum_{i=1}^{k+1} \frac{1}{i(i+1)} = \frac{(k+1)}{(k+1)+1} = \frac{k+1}{k+2}$$

④ Proof

your proof should look like this

$$\sum_{i=1}^{k+1} \frac{1}{i(i+1)} = \underset{\text{use algebra}}{\overset{\text{use step ②}}{\cdots\cdots\cdots}} = \frac{k+1}{k+2}$$

STEP 2

$$\sum_{i=1}^{k+1} \frac{1}{i(i+1)} = \sum_{i=1}^{k} \frac{1}{i(i+1)} + \frac{1}{(k+1)(k+1+1)} = \frac{k}{k+1} + \frac{1}{(k+1)(k+2)}$$

$i = 1, 2, 3, \ldots, k, \overline{k+1}$

$$= \frac{k}{k+1} \cdot \frac{k+2}{k+2} + \frac{1}{(k+1)(k+2)}$$

$$= \frac{k^2+2k+1}{(k+1)(k+2)} = \frac{(k+1)(k+1)}{(k+1)(k+2)} = \frac{k+1}{k+2}$$

Matches the target

∴ By the principle of Mathematical induction $\sum_{i=1}^{n} \frac{1}{i(i+1)} = \frac{n}{n+1}$; $n \geq 1$

2. What does the following function return? Can you prove that by mathematical induction?

```
MysteryFunction (positive integer n)

    i = 1
    j = 5
    while i ≠ n do
        j = 2j - 3
        i = i + 1
    end while

return j
```

Funct. non recursive formula

n	Funct code	$j_n = 2^n + 3$
1	5	$2^1+3=5$
2	$2(5)-3=7$	$2^2+3=7$
3	$2(7)-3=11$	$2^3+3=11$
4	$2(11)-3=19$	$2^4+3=19$

The function returns $j_n = 2^n + 3$; $n \geq 1$

① Base case, $n=1$ (verify)

code: For $n=1$ while loop does not run because the condition while $(i \neq n)$ is while $(1 \neq 1)$ and False. The code returns $j_1 = 5$

formula: $j_1 = 2^1 + 3 = 5$

the Base case holds

② Assume that the following is true for $n=k$

The function returns $j_k = 2^k + 3$

③ Target for $n=k+1$

The function returns $j_{k+1} = 2^{k+1} + 3$

④ Proof your proof should look like this: for $n=k+1$ the function returns use step 2 use code use algebra ...... $2^{k+1}+3$

matches the target

When $n=k+1$ is passed to the function, in the last iteration of the while loop j_{k+1} is calculated as: $j_{k+1} = 2 \cdot \underbrace{j_k}_{\text{STEP 2}} - 3$

$= 2 \cdot (2^k + 3) - 3 = 2 \cdot 2^k + 2 \cdot (3) - 3 = 2^{k+1} + 3$

∴ By the principle of Mathematical Induction

the function returns $j_n = 2^n + 3$; $n \geq 1$

3. Use mathematical induction to show that `MysteryFunction(n)` returns $9n + 47 \cdot 4^{n-1} + 23$.

```
MysteryFunction (positive integer n)

    i = 1
    j = 79
    while i ≠ n do
        i = i + 1
        j = 4j - 27i - 33
    end while

return j
```

① Base case, $n = 1$ (verify)

code: For $n=1$ the condition while $(i \neq n)$ is while $(1 \neq 1)$ and false.
The code returns $j_1 = 79$

formula: $j_1 = 9(1) + 47 \cdot (4)^{1-1} + 23 = 9 + 47(1) + 23 = 79$

The Base case holds

② Assume that the following is true for $n = k$

The function returns $j_k = 9k + 47 \cdot 4^{k-1} + 23$

③ Target for $n = k+1$

The function returns $j_{k+1} = 9(k+1) + 47 \cdot 4^{k+1-1} + 23$

$= 9k + 9 + 47 \cdot 4^k + 23$ simplify

$= 9k + 47 \cdot 4^k + 32$

④ Proof

when $n = k+1$ is passed to the function, in the last iteration of the while loop j_{k+1} is calculated as:

$j_{k+1} = 4 j_k - 27 i - 33 = 4(9k + 47 \cdot 4^{k-1} + 23) - 27 i - 33$

From step 2 we have $j_k = 9k + 47 \cdot 4^{k-1} + 23$.

the condion $i \neq n$ is $i \neq k+1$ meaning $i = k$, then incremented to $i = k+1$ in the next line

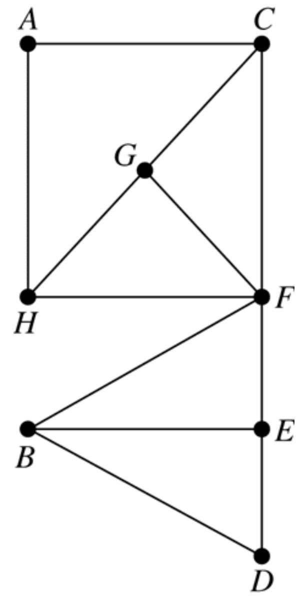
$= 36k + 4 \cdot 47 \cdot 4^{k-1} + 92 - 27(k+1) - 33$

$= 36k + 47 \cdot 4^{1+k-1} + 92 - 27k - 27 - 33 = 9k + 47 \cdot 4^k + 32$

∴ By The principle of Math. Induct. the function returns $j_n = 9n + 47 \cdot 4^{n-1} + 23$; $n \geq 1$

Graphs and Trees

1. Let G = (V, E) be graph in the figure. How many paths are there from A to D? How many of these paths have length 6?

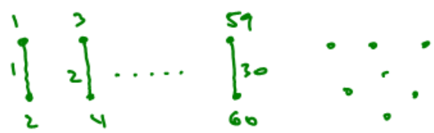


(# of paths A→F)·(# of paths F→D) = 6·4 = 24

	AF length	FD length		
3+3=6	3	3	→ 2(2) = 4	+ = 8
4+2=6	4	2	→ (2)(2) = 4	

2. Consider a simple[1] graph G = (V, E) with 30 edges.

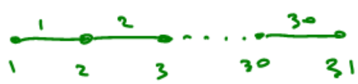
a) What is the maximum number of vertices that G can have?



← there could be many isolated verticies

∴ no upper limit

b) If G is connected, what is the maximum number of vertices that G can have?



∴ |V| = 31

c) What is the minimum number of vertices that G can have?

the graph with n verticies can have up to $\binom{n}{2}$ edges

n	max number of verticies $\binom{n}{2}$	
6	$\binom{6}{2} = \frac{6(5)}{2} = 15$	can not fit 30 edges
7	$\binom{7}{2} = \frac{7(6)}{2} = 21$	
8	$\binom{8}{2} = \frac{8(7)}{2} = 28$	
9	$\binom{9}{2} = \frac{9(8)}{2} = 36$	← 30 edges fits!

n = 9

[1] A simple graph is an unweighted, undirected graph containing no loops or multiple edges between two vertices (do not confuse "loop" with "cycle").

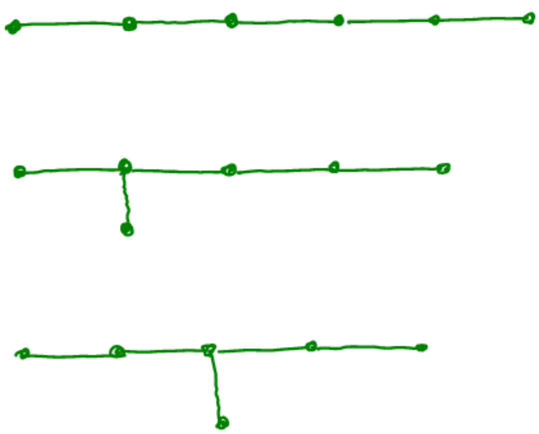
3. Give an example of a connected graph G where removing any edge results in a disconnected graph. Can you generalize this result?

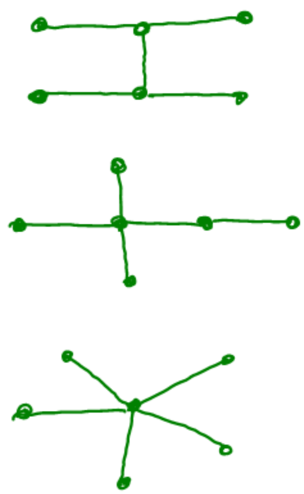
one example:

↳ it means that the graph has no cycle.
+ connected

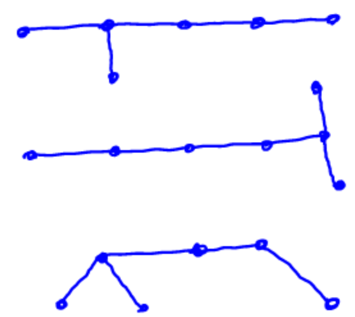
} = tree

4. Draw all non-isomorphic trees on six vertices.





examples of isormophic trees



5. Give an example of an undirected graph G = (V, E) where |V| = |E| + 1 but graph is not a tree.

if tree then $|V| = |E| + 1$

In both cases $|V| = |E| + 1$
$7 = 6 + 1$

But

if $|V| = |E| + 1$ → tree

→ not a tree

6. If a tree has three vertices of degree 2, four vertices of degree 3, four vertices of degree 4, and two vertices of degree 5, how many vertices with degree 1 does it have?

# of vertices	deg type
3	2
4	3
4	4
2	5
x	1

$+ = |V| = 13 + x$

← it means 

$|V| = |E| + 1$ works for any tree

$13 + x = |E| + 1$

$|E| = 12 + x$

$\sum_{i=1}^{|V|} \deg(v_i) = 2|E|$ works for any graph, so it works for tree too

$3(2) + (4)(3) + 4(4) + 2(5) + x(1) = 2(12 + x)$

$44 + x = 24 + 2x$

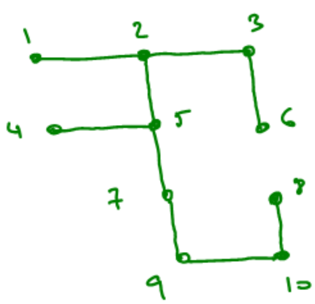
$44 - 24 = 2x - x$

$x = 20$

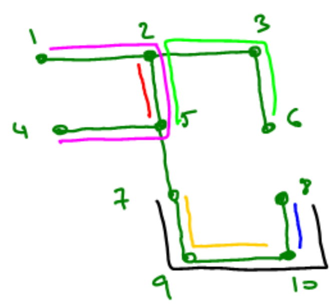
$\therefore$ There are 20 vertices with degree 1

7. Let T = (V, E) be a tree with 10 vertices. How many distinct paths are there in T?

An example of tree 10 vertices



Examples of distinct paths



tree has no cycle → there is a single path between any vertices

In a tree with n vertices, there are $\binom{n}{2}$ different pairs of vertices and therefore $\binom{n}{2}$ distinct paths

$\therefore \binom{10}{2} = \frac{10(9)}{2} = 45$ distinct paths

If all of the problems can't be done in the lab period, students can work on them on their own.